Tut 1

Ex. 7.6 No. 50

$$\int \frac{dx}{9+3x^2} = \frac{1}{3}\int \frac{dx}{3+x^2}$$

$$3+x^2 = \boxed{3}\left(1+\frac{x^2}{3}\right)$$

$$= 3\left(1+\left(\frac{x}{\sqrt{3}}\right)^2\right)$$

$$= \frac{1}{3}\cancel{\int}\frac{1}{\sqrt{3}}\arctan\left(\frac{x}{\sqrt{3}}\right)+C$$

$$= \frac{\sqrt{3}}{9}\arctan\left(\frac{x}{\sqrt{3}}\right)+C$$

Ex. 8.1

Q25 $\displaystyle\int \frac{\boxed{dy}}{\sqrt{e^{2y}-1}} = \int \frac{du}{u\sqrt{u^2-1}} = \operatorname{arcsec} u + C$

$= \operatorname{arcsec} e^y + C$

$e^{2y}-1 = (e^y)^2 - 1$

$u = e^y$

$du = e^y\,dy$

$dy = \frac{1}{e^y}\,du$

$= \frac{1}{u}\,du$

Q40 $\displaystyle\int \frac{\sqrt{x}}{1+x^3}\,dx = \int \frac{2}{3}\cdot\frac{1}{1+u^2}\,du$

Hint: $u = x^{3/2}$ $\qquad = \frac{2}{3}\arctan u + c$

$1+x^3 = 1+\left(x^{\frac{3}{2}}\right)^2$ $\qquad = \frac{2}{3}\arctan x^{3/2} + c$

$u = x^{3/2}$

$du = \frac{3}{2}\boxed{\sqrt{x}\,dx}$

$\sqrt{x}\,dx = \frac{2}{3}\,du$

Q34 $\int e^{z} \pm e^{z}\, dz$

$$= \int e^{z} e^{e^{z}}\, dz \qquad u = e^{z} \qquad du = e^{z}\, dz$$

$$= \int e^{u} \cancel{+C}\, du$$

$$= e^{u} + C$$

$$= e^{e^{z}} + C$$

$-\csc\theta$

$\uparrow$ int.

$\cot\theta\csc\theta$

$\uparrow$

$\frac{\cos\theta}{\sin^2\theta}\ \frac{1}{\sin\theta}$

$-\cot\theta$

$\uparrow$ int.

$\csc^2\theta$

$\uparrow$

Q38 $\int \frac{d\theta}{\cos\theta - 1} = -\int \frac{\cos\theta + 1}{\sin^2\theta}\, d\theta = -\int \frac{\cos\theta}{\sin^2\theta}\, d\theta - \int \frac{1}{\sin^2\theta}\, d\theta$

$$\frac{1}{\cos\theta - 1} \cdot \frac{\cos\theta + 1}{\cos\theta + 1} = \frac{\cos\theta + 1}{\cos^2\theta - 1} = -\frac{\cos\theta + 1}{\sin^2\theta}$$

$\cos^2\theta + \sin^2\theta = 1$

Q26

$$\int \frac{6\,dy}{\sqrt{y}\,(1+y)} = \int \frac{6\cdot 2\,du}{(1+u^2)} = 12\arctan u + c$$

$$= 12\arctan\sqrt{y} + c$$

$1+(\sqrt{y})^2$

$$\frac{f(y)}{\sqrt{y}}$$

$$f(y) = \frac{6}{1+y}$$

$$d\sqrt{x} = \frac{1}{2\sqrt{x}}\,dx$$

$$u = \sqrt{\cancel{x}\,y}$$

$$du = \frac{1}{2\sqrt{\cancel{x}\,y}}\,d\cancel{x}\,y$$

Q 32

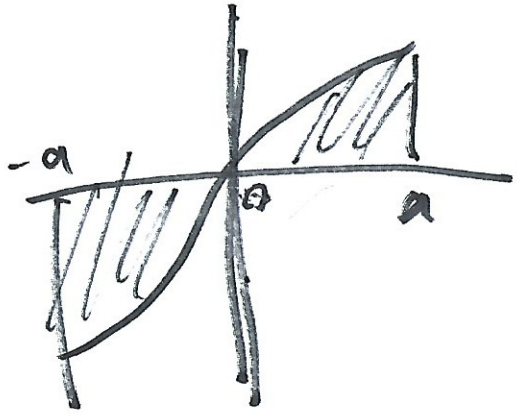
$$\int_{-1}^{1} \sqrt{1+x^2}\,\sin x \, dx = 0$$

$u = x^2$

$du = 2x\,dx$ (Not good)

$$\int_{-a}^{a} f(x)\,dx = 0$$

if f is an odd function.



Odd func. $f(-x) = -f(x)$

$$f(x) := \sqrt{1+(-x)^2}\,\sin(-x) = \sqrt{1+x^2}\,(-\sin x) = -f(x)$$

$$\int_{-a}^{a} f(x)\,dx = \int_{0}^{a} f(x)\,dx + \int_{-a}^{0} f(x)\,dx$$

$u = -x$

$du = -dx$

$x = 0$

$\Rightarrow u = 0$

$x = -a$

$u = a$

$$= \int_{0}^{a} f(x)\,dx + \int_{a}^{0} -f(-u)\,du$$

$$= \int_{0}^{a} f(x)\,dx + \int_{a}^{0} f(u)\,du$$

$f(-u) = -f(u)$

$$= \int_{0}^{a} f(x)\,dx - \int_{0}^{a} f(u)\,du$$

$-f(-u) = f(u)$

$$= \int_{0}^{a} f(x)\,dx - \int_{0}^{a} f(x)\,dx$$

$$= 0.$$

Q9

$$\int \frac{dz}{e^z + e^{-z}} \cdot \frac{e^z}{e^z}$$

$$= \int \frac{e^z \, dz}{e^{2z} + 1}$$

$$e^{2z} + 1 = (e^z)^2 + 1$$

$$u = e^z \Rightarrow du = \underline{\underline{e^z \, dz}}$$

$$= \int \frac{1}{u^2 + 1} \, du$$

$$= \arctan u + C$$

$$= \arctan e^z + C$$